SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

Code of Conduct

I ATHAVAN (192524036)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ATHAVAN K

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1- AT-1 Short Answer Test	Name :: Athavan K Reg No :: 192524036 Course code : CSA1507 Course :: cloud computing and Big Data Analytics

1 cloud computing :-

* where it provides the service according to the pay-use demand. If you want a storage base for storing your files, you can pay according to your need and get the storage.

* It delivers computing resource (or) service like servers, storage, databases, networking and softwares

* Instead of buying and maintaining physical data centers, users rent access to these resource from providers like Amazon web, Microsoft Azure (or) Google cloud.

characteristics of cloud computing :

(i) on Demand self services : we can buy services as the you wish without any direct contact.

(ii) Broad network Access : we can access the cloud services with our networking gadgets like Laptops, Smartphone, Pc's ...etc.
Eg) If, a student access services through Laptop in college hours, he can access the service in through mobiles. ~~phone~~.

(iii) Rapid elasticity : If a person uses the cloud platform for his shop. During Festival time the customers might increase so, in that case he can increase the cloud service according to the sales without crashing. And after the Festival time he can reduce the cloud services.

(iv) Measured services:

* There are four types of measured service

* Infrastructure as a service (IaaS)

* Platform as a service (PaaS)

* Software as a service (SaaS)

* Anything as a service (XaaS)

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2 Evolution of cloud computing:

As we the modern computing doesn't jump from nowhere, It has Evolved throughout the years. It undergone several stages.

(i) Hardware Evolution:

* At the ~~yearly~~ early years of computing the hardwares cost high, like microchips are macro in size and the computing is so large.

* Moving on to the years the hardwares become small with large size of memory and cost has been reduced.

* It helps users to the buy computers and became popular.

5 These IaaS, PaaS, SaaS, and XaaS are comes under measured

(ii) Networking :-

* At early years we use cards with holes for computing.

* Moving on to the years the computers are modernised into with the subject computer networks.

* Networks made easier to communicate between computers

(iii) Distributed computing :-

* Through the networks, the computers can share the data. So, the works for a single users divided into many and to other computers.

(iv) Internet :

* Due to internet evolution, the computers can be connected to world wide and can share the files.

(v) Internet as a software :

* Due to evolution of internet, then creation of Application becomes common and shared among computers.

(vi) Modern computing :- Due to these changes virtualization happened as it runs without pay use demand and It is part of computing.

aas Paas, Saas, and ...

4 Role of a Data centers :

what is Data centers :

* Datacentres are system that are House servers, storage system, networking equipments and security infrastructure that are required for Deliversing a cloud services.

* They are more (or) less a physical servers.

Internet → Datacentres → Storage → Services.

Roles :

* It stores and processes large amount of data

* It Hosts cloud applications and services

* It provides availability and reliability through Back-up-system.

* It ensures Data integrity and security by encryption

* It is helpful in providing services by adding more services when needed.

5 These Iaas, Paas, Saas, and Xaas are comes under measured services.

Iaas:-

* It is called Infrastructure as a service.
* where it provides storages, networking, Data security and Data Integrity.
* The users has high control over in it.
* Ex) where Amazon, Google cloud provide those things.

Paas:-

* It is called as Platform as a service.
* where it provides platform, to the users. where Developers can Develop a software.
* The user has mid level control over in it.
* Ex) Google search Engine.

Saas:-

* It is called a ~~service~~ Software as a service
* where it provides software (or) Direct Application to the user.
* The user has very low level control in it.
* Ex) Google Docs.

SaaS:-

* It is called software as a service

* The user can what resource he want

* The send of ~~~~ according to the service

* (Ex) Microsoft 365

2 Role of server virtualization:-

* where it act as a physical operating system with the help of hypervisor.

* ~~It~~ There can be multiple virtual machine in a single system according to its capacity.

* According to the work capacity the VM can be allocated by the user.

* Each VMs are isolated with each other, if any one failed all other can run without ~~any~~ interruption

virtualization	cloud computing
* It can perform any tasks assigned to it without physical servers.	* It is physical infrastructure located in a place.
* It perform the tasks	* where as It provides the services.
* There can more VMs created in a OS System	* where it create the VMs.
* where it use the service	* It provides it.